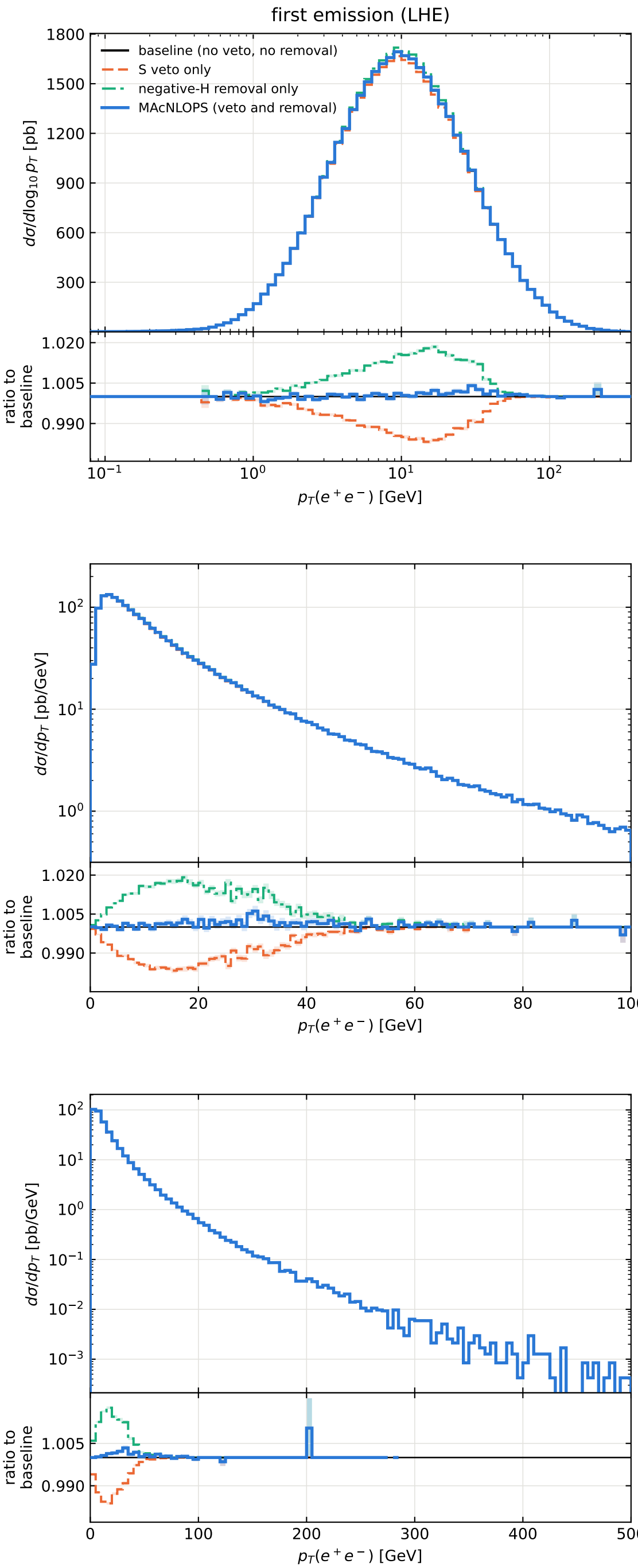
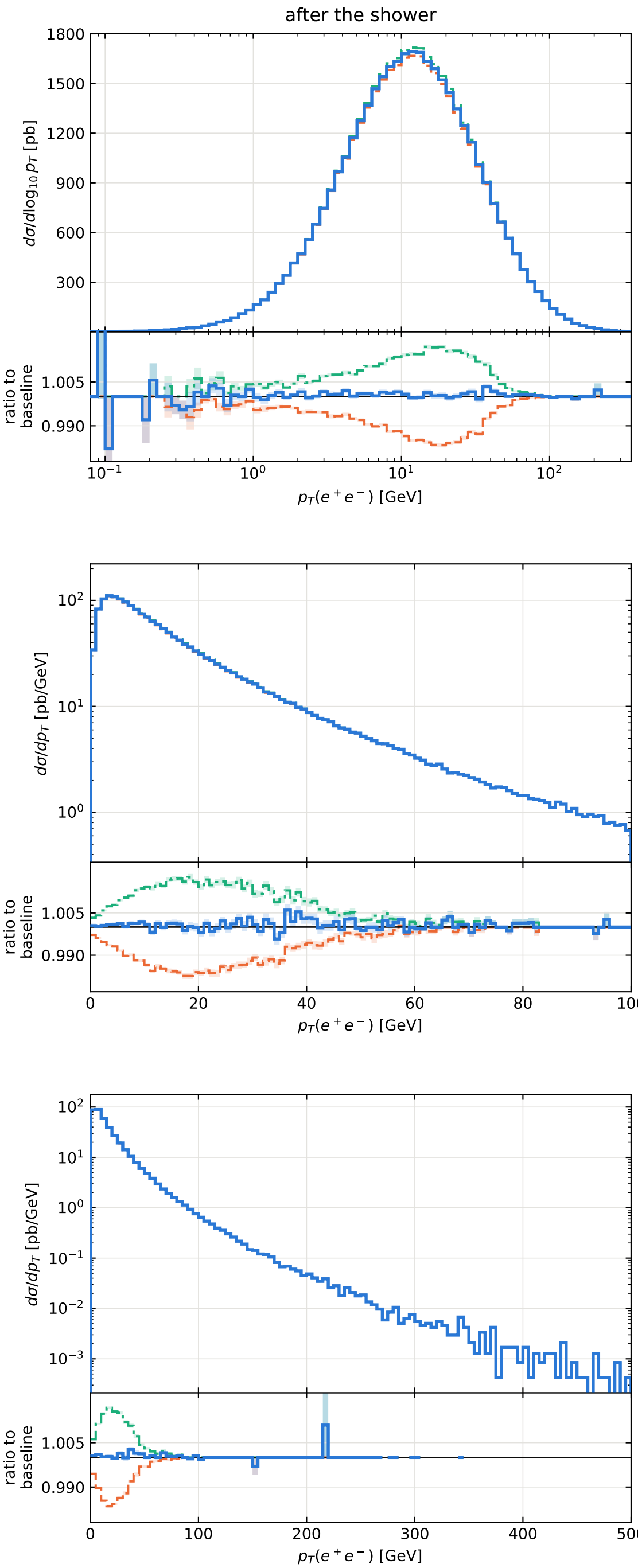
$pp \to e^+e^-$ (DY, $m_{ee} > 60$ GeV): 1000000 events, 13 TeV, NNPDF4.0 NLO, common-scale MC@NLO-Δ source

Shower: Pythia 8.318 from the first-emission endpoints, without MPI, hadronisation, primordial k_T, QED or top/Higgs decays; 50 split jobs

Totals: baseline 1932 pb (+0.00%) S veto only 1913 pb (-1.02%) negative-H removal only 1953 pb (+1.09%) MAcNLOPS 1934 pb (+0.07%)

Ratio bands: correlated statistical uncertainty (per event at the first emission, batch means over the split shower jobs after the shower), where the baseline is known to 25%.



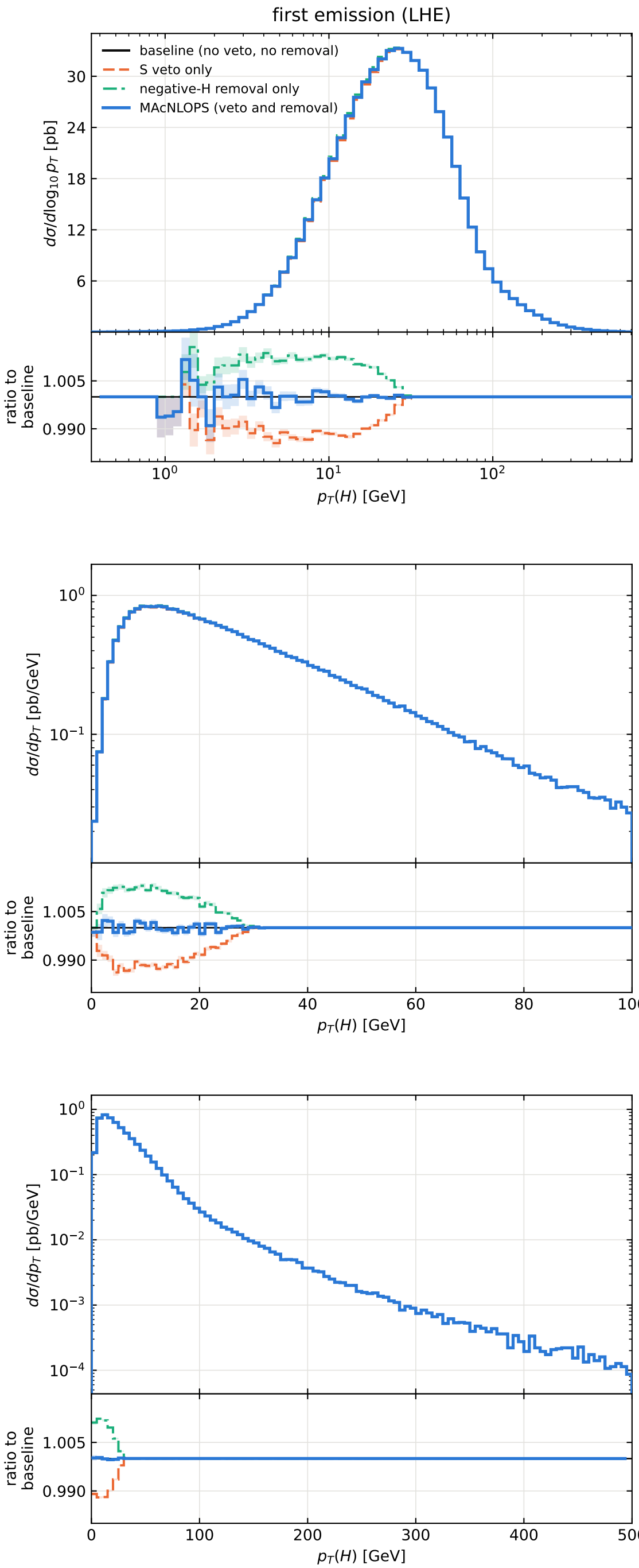


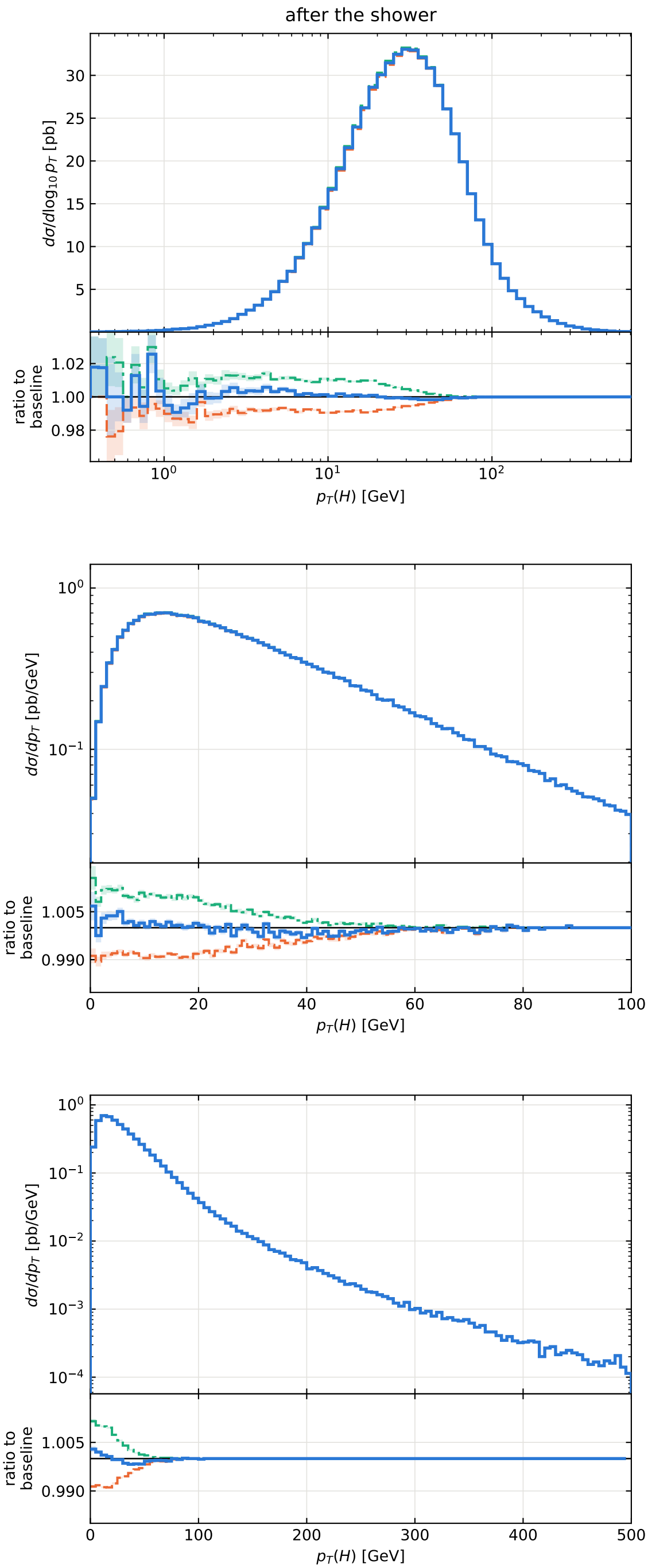
gg→H (HEFT): 1000000 events, 13 TeV, NNPDF4.0 NLO, common-scale MC@NLO-Δ source

Shower: Pythia 8.318 from the first-emission endpoints, without MPI, hadronisation, primordial k_T, QED or top/Higgs decays; 50 split jobs

Totals: baseline 30.72 pb (+0.00%) S veto only 30.55 pb (-0.55%) negative-H removal only 30.89 pb (+0.54%) MAcNLOPS 30.72 pb (-0.00%)

Ratio bands: correlated statistical uncertainty (per event at the first emission, batch means over the split shower jobs after the shower), where the baseline is known to 25%.





$pp \to t\bar{t}$: 1000000 events, 13 TeV, NNPDF4.0 NLO, common-scale MC@NLO-Δ source

Shower: Pythia 8.318 from the first-emission endpoints, without MPI, hadronisation, primordial k_T, QED or top/Higgs decays; 50 split jobs

Totals: baseline 636.8 pb (+0.00%) S veto only 573.7 pb (-9.90%) negative-H removal only 695.7 pb (+9.26%) MAcNLOPS 632.7 pb (-0.64%)

Ratio bands: correlated statistical uncertainty (per event at the first emission, batch means over the split shower jobs after the shower), where the baseline is known to 25%.

